

Anytime Statistical Spending for Menu Audits: Change-Control-Safe Confidence Budgets for Expanding Log Certificates

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Abstract

Synthesis 25 gives a conservative finite-sample recipe that turns log frequencies for a *fixed* discrete projection into a simultaneous (menu-level) total-variation certificate. Real deployments are not one-shot: menus expand, audits repeat, and drift monitoring triggers incremental re-certification.

This note centralizes a minimal, referee-grade *anytime spending* rule: how to allocate a single published statistical failure budget δ_{stat} across an unbounded sequence of audit events (and across expanding menus) without invalidating earlier certificates. The goal is non-redundancy: mechanism and contract papers can cite this note rather than re-deriving optional-stopping or “menu growth” bookkeeping.

1 Problem: certificates that must survive time

Many papers in this archive publish claims of the form

$$\text{TV}(P_{k,\mu}, P_{k',\mu}) \leq \tau_\mu$$

for discrete projections X_μ audited from logs. Synthesis 25 supplies the per-audit certificate surface.[?]

Operationally, audits occur at times $t = 1, 2, \dots$ (weekly, per-release, or when monitors fire). At time t , the deployment may: (i) add new keys/policies to an existing menu, (ii) add a new projection (new ENF-ID), (iii) repeat an audit with fresh logs, or (iv) publish drift evidence that triggers a new receipt under the plan/state equivalence rules.[?, ?]

A naive reuse of δ per audit silently violates the meaning of the published statistical confidence. This note fixes the bookkeeping with a single device: *spend* δ_{stat} over time.

2 Spending schedules

Definition 1 (Audit event). *An audit event A_t is the act of publishing (or internally producing) a finite set of statistical certificates at time t . Each event consumes a chosen failure probability budget $\delta_t \in (0, 1)$.*

Definition 2 (Spending schedule). *A spending schedule is any sequence $(\delta_t)_{t \geq 1}$ such that*

$$\sum_{t=1}^{\infty} \delta_t \leq \delta_{\text{stat}}.$$

We treat the schedule name (e.g., `geom-1/2` or `zeta2`) as a published parameter.

Two simple schedules (both easy to publish and implement):

- **Geometric:** $\delta_t = \delta_{\text{stat}} 2^{-t}$.
- **Zeta(2):** $\delta_t = \delta_{\text{stat}} \frac{6}{\pi^2} \frac{1}{t^2}$.

Both satisfy $\sum_t \delta_t \leq \delta_{\text{stat}}$.

Lemma 1 (Anytime validity by a union bound). *Suppose audit event A_t is designed so that its published certificates hold with probability at least $1 - \delta_t$. If (δ_t) is a spending schedule, then with probability at least $1 - \delta_{\text{stat}}$, all certificates published across all times hold simultaneously.*

Proof. Apply a union bound over events: $\Pr[\exists t : A_t \text{ fails}] \leq \sum_t \delta_t \leq \delta_{\text{stat}}$. $\square$

Remark 1 (Why this is enough for this archive). *This note is intentionally conservative. More powerful anytime-valid tools (e-values, confidence sequences) exist, but Lemma ?? is short, transparent, and compatible with the receipt/line-item publication model. When a paper needs stopping-safe drift tests for state witnesses, the state-series owns that story; this note only standardizes how to publish log-audit confidence over time.[?]*

3 Expanding menus inside an audit event

Within an event t , the deployment may certify a *finite* set of menu items and class pairs. Write $\mathcal{I}_t$ for the set of audited pairs (k, μ) at time t . A simple in-event allocation is Bonferroni:

$$\delta_{t,(k,\mu)} := \delta_t / |\mathcal{I}_t|.$$

Then Corollary (menu-level radii) from Synthesis 25 applies inside t with δ_t as the event-level confidence.[?]

Corollary 1 (Anytime-valid menu growth). *Assume each audit event t certifies its (finite) scope $\mathcal{I}_t$ with failure probability at most δ_t . Under any spending schedule, with probability at least $1 - \delta_{\text{stat}}$, every menu item ever certified remains valid, even as the menu grows over time.*

Practical interpretation. A deployment may publish a new key-class, add a new projection, or expand a certified menu without “resetting” statistical meaning, as long as it spends additional δ_t and publishes that spend.

4 Receipt-facing publication fields (minimal)

This archive’s receipts already separate *claim surfaces* (ENF-ID, TW-ID, optional state-decl id) from *evidence objects*. [?, ?] This note proposes a minimal set of fields for evidence objects that rely on log-audit spending:

- δ_{stat} (menu-level statistical confidence budget),
- **spending schedule id** (e.g., `geom-1/2` or `zeta2`) and its parameter(s),
- **event index** t (or an event nonce) and the spent value δ_t ,
- **in-event allocation rule** (e.g., Bonferroni) and per-item sample sizes.

The rest of the per-projection evidence surface (projection id, menu scope, empirical distances, derived bounds) is owned by Synthesis 25, and the interlock to receipts and bundles is exercised concretely in Synthesis 17.[?, ?]

5 Change control and identity

Spending does *not* change claim identity. If ENF-ID / TW-ID / state-decl id are unchanged, an audit event can be published as a new evidence object referenced by the same claim. If any identity field changes, Synthesis 18’s plan/state taxonomy should treat the outcome as a new claim object (or as an explicit coarsening) and require fresh evidence.[?]

6 Research warranted (short list)

- **E-value unification:** connect the state-series stopping-safe drift tests to the log-audit spending surface so deployments can publish a single “monitoring confidence” parameter without mixing regimes.
- **Adaptive projection discovery:** when the projection X is discovered from logs (feature selection), the effective multiplicity can dwarf a fixed menu size; bridging to Certified A/C selection wrappers is a natural next note.[?]
- **Power-aware spending:** schedules that allocate more δ to high-impact events (e.g., major releases) while retaining a simple publication rule.

References

- [1] Anonymous. Discrete Equalization Audits from Logs: Menu-Level Finite-Sample TV Certificates. Synthesis 25, The-One-True-Archive (2026).
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- [3] Anonymous. Receipt Line Items for Anonymity Claims: A Minimal Tuple Schema for Published Budgets. Synthesis 12, The-One-True-Archive (2026).
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